
Integrable Systems — Exercise Sheet 16

The thermodynamic limit

Variant: Questions only

Conventions, as in Lecture 16. The chain: $H = J \sum_n (\mathbf{S}_n \cdot \mathbf{S}_{n+1} - \frac{1}{4})$ on a ring of N sites, $J > 0$ ($J = 1$ in computations); $\theta_n(\lambda) = 2 \arctan(2\lambda/n)$; the logarithmic Bethe equations $N\theta_1(\lambda_j) - \sum_l \theta_2(\lambda_j - \lambda_l) = 2\pi I_j$, $I_j \in \mathbb{Z} + \frac{N-M+1}{2}$; the counting function $\zeta(\lambda) = \theta_1(\lambda) - \frac{1}{N} \sum_l \theta_2(\lambda - \lambda_l)$; $E = -\sum_j 2J/(4\lambda_j^2 + 1)$; $a_n = \theta'_n/2\pi$; and $\hat{f}(\omega) = \int f(\lambda) e^{i\omega\lambda} d\lambda$. The gas: $H = -\sum_j \partial_j^2 + 2c \sum_{i<j} \delta(x_i - x_j)$, $c > 0$, on a ring of length L ; $\theta(k) = 2 \arctan(k/c)$; $k_j L + \sum_l \theta(k_j - k_l) = 2\pi I_j$; $K(k) = 2c/(k^2 + c^2)$; Lieb's equation $\rho(k) = \frac{1}{2\pi} + \frac{1}{2\pi} \int_{-q}^q K(k - k') \rho(k') dk'$, with $n = \int \rho$, $E/L = \int k^2 \rho$, $\gamma = c/n$ and $E/N = n^2 e(\gamma)$.

Exercise 1 (the antiferromagnet by Fourier transform). (i) Show that $\hat{a}_n(\omega) = e^{-n|\omega|/2}$. (ii) Transform the ground-state equation $\rho + a_2 * \rho = a_1$ to find $\hat{\rho}(\omega) = 1/(2 \cosh(\omega/2))$. Lecture 16 states that this is the transform of $1/(2 \cosh \pi\lambda)$: prove it. (Parts (i) and (ii) use the residue theorem and the uniqueness of the Fourier transform, neither among Lecture 0's prerequisites; Lecture 16 uses both.) (iii) Show that $\int_{-\infty}^{\lambda} \rho = \frac{1}{\pi} \arctan e^{\pi\lambda}$, and find the fraction of the ground state's roots with $|\lambda| < 1$ in the thermodynamic limit.

Exercise 2 (the gas's counting function, and the sign). (i) For the gas put $z(k) = k + \frac{1}{L} \sum_l \theta(k - k_l)$, so that $Lz(k_j) = 2\pi I_j$. Assuming, as Lecture 16 does, that roots and holes fill out densities ρ , ρ_h , derive $\rho + \rho_h = \frac{1}{2\pi} + \frac{1}{2\pi} \int K(k - k') \rho(k') dk'$, and Lieb's equation for the ground state. (ii) Show that z is strictly increasing for every set of real roots. Is ζ ? (iii) Write both models as $Lp_0(\lambda_j) + \sum_l \varphi(\lambda_j - \lambda_l) = 2\pi I_j$ and find the two kernels $K = \varphi'$.

Exercise 3 (strong coupling). (i) For N free fermions on the ring, N odd, with momenta $2\pi I/L$, $I = -\frac{N-1}{2}, \dots, \frac{N-1}{2}$, show that $E/N = \frac{\pi^2 n^2}{3} (1 - \frac{1}{N^2})$, which tends to Lecture 16's Tonks–Girardeau value $\pi^2 n^2/3$. (ii) For $\gamma \gg 1$, use $K(k - k') \approx 2/c$ on the sea to show that $\rho \approx \frac{1}{2\pi} (1 + \frac{2}{\gamma})$, that $q \approx \pi n/(1 + \frac{2}{\gamma})$, and that $e(\gamma) = \frac{\pi^2}{3} (1 - \frac{4}{\gamma} + O(\gamma^{-2}))$. Compare q with Lecture 11's ring lengthened by $2N/c$.

Exercise 4 (Lieb's equation by quadrature). (Computational.) Set $c = 1$. (i) Solve Lieb's equation at $q = 1$ by the Nyström method of Lecture 16's tutorial with 40 Gauss–Legendre points and give n , E/L , γ and e . How far is n from this value with 5, 10 and 20 points? (ii) By Hellmann–Feynman the interaction energy is $c \partial E / \partial c$ at fixed N , L . Show that the kinetic and interaction energies are equal exactly where $d \ln e / d \ln \gamma = \frac{1}{2}$, and find that γ and $e/2$ there. (iii) Plot $e(\gamma)$ for $0.02 \leq \gamma \leq 300$ on a logarithmic axis, with the two series of Lecture 16's reading, and give $e(0.1)$ and $e(40)$ beside them.

Exercise 5 (the antiferromagnet's roots). (Computational.) Take $N = 100$, $M = 50$ and $I_j = -\frac{49}{2}, \dots, \frac{49}{2}$. (i) Solve the logarithmic Bethe equations by damped Newton, and give E_0/N . (ii) Plot the local density $2/[N(\lambda_{j+1} - \lambda_{j-1})]$ against $1/(2 \cosh \pi\lambda)$, and give the worst relative deviation five roots in from each end. (iii) Lecture 16's reading predicts $E_0(N) = Ne_0 - \pi c v/(6N) + o(N^{-1})$ with $v = \pi J/2$. Compute $-(E_0 - Ne_0) 6N/\pi$ at $N = 100, 200, 400$, and estimate the central charge c .

Exercise 6 (two roots, one quantum number). (Computational.) Lecture 16's tutorial meets two roots of $N = 22$, $M = 2$ that share $I_1 = I_2 = \frac{19}{2}$. (i) Solve the two logarithmic

Bethe equations with $I_1 = I_2 = \frac{19}{2}$ by Newton from (2.23, 2.37), to 12 decimals. (ii) Evaluate ζ' at both roots and at their midpoint. Which hypothesis of Lecture 16's vacancy count fails, and which of its statements survive? (iii) Build Lecture 10's two-magnon wavefunction on them, $\psi(x_1, x_2) = e^{i(k_1 x_1 + k_2 x_2)} + A_{21} e^{i(k_2 x_1 + k_1 x_2)} \quad (x_1 < x_2)$, with $e^{ik_j} = \frac{\lambda_j + i/2}{\lambda_j - i/2}$ and $(\lambda_1 - \lambda_2 + i)A_{21} = \lambda_1 - \lambda_2 - i$, and check that it is an eigenvector of H ($J = 1$).

Exercise 7 ★. *Why Lieb's equation has one positive solution.* Fix $c > 0$ and $q > 0$, and let $(\mathcal{K}f)(k) = \frac{1}{2\pi} \int_{-q}^q K(k - k') f(k') dk'$ on continuous functions on $[-q, q]$ with the sup norm. (i) Show that $\|\mathcal{K}\| = \frac{2}{\pi} \arctan \frac{q}{c} < 1$, and deduce that Lieb's equation has exactly one continuous solution, $\rho = \sum_{m \geq 0} \mathcal{K}^m \frac{1}{2\pi}$. (ii) Show that $\rho \geq \frac{1}{2\pi}$. (iii) For $q < q'$ show that $\rho_{q'} \geq \rho_q$ on $[-q, q]$, and deduce that $n(q)$ is strictly increasing, so that $\gamma = c/n$ is strictly decreasing in q : the step Lecture 16's proof of Lieb's theorem uses.